KERALA POLYTECHNIC · REVISION 2026

Course 2001B | Semester II | Theory · 4 Credits | 3:1:0:0

Mathematics for Electrical Sciences

Complex Numbers · Vector Algebra · Integral Calculus · Differential Equations & Laplace Transforms

Complete handbook for the Kerala Diploma Revision 2026 curriculum. Covers all four modules with deep explanations, worked examples, interactive calculators, Malayalam support, and exam-oriented practice.

PROGRAMME	Biomedical Engg, ECE, EEE, EE, Electronics Engg, Embedded Systems, Electronics & Computer Engg, EV Technology, IC Design, Instrumentation, Micro Electronics, Renewable Energy
COURSE TYPE	Theory
CONTACT HOURS	60 (L:45 + T:15)
SELF-LEARNING	60 hours (suggested)
CIE / ESE	40 / 60

MODULE I	Complex Numbers (14 h)
MODULE II	Vector Algebra (15 h)
MODULE III	Integral Calculus (15 h)
MODULE IV	Differential Equations & Laplace Transforms (16 h)

📄 Official Source Declaration

Source: Kerala Polytechnic Revision 2026 Syllabus — Course 2001B (Mathematics for Electrical Sciences). This handbook is an educational study aid developed for students. It is not an official publication of the examining body.

⚠️ **Verified Inconsistencies:** The PDF shows "Mathematics for Electrical Sciences" as both the course title and the textbook author (SITTTR Kalamasser). This appears to be a formatting artifact — the textbook is likely the same SITTTR publication used for 2001A, with a different title. The assessment methodology table is garbled but has been interpreted consistently with the data provided. All other data reconcile.

📊 Course Dashboard

60

CONTACT HOURS

4

CREDITS

40+60

CIE / ESE

4

MODULES

📌 Prerequisites

Fundamentals of Engineering Mathematics I (Course 1002) — Trigonometry, Determinants, Differentiation.

Teaching & Assessment Scheme

Teaching Scheme	Self-Learning/Week	Total Hours	Credits	CIE	ESE
3:1:0:0 (L:T:P:R)	5.5 h	60	4	40	60

Module-wise Instructional Hours

Module	Title	Lecture (L)	Tutorial (T)	Total
I	Complex Numbers	10	4	14
II	Vector Algebra	12	3	15
III	Integral Calculus	11	4	15
IV	Differential Equations & Laplace Transforms	12	4	16
Total			**60**	

🧭 How to Use This Handbook

1

Understand

Read each module's explanation, formulae, and worked examples.

2

Visualise

Study the SVG diagrams and interactive calculators.

3

Apply

Solve the module-wise questions and try the model paper.

4

Revise

Use the Quick Revision cards and glossary before exams.

🎯 Course Outcomes

CO1

Apply concepts of complex numbers to solve problems. (Bloom: Apply)

CO2

Identify and apply vector algebra techniques to solve problems involving scalar and vector quantities. (Bloom: Apply)

CO3

Use integration techniques to solve basic engineering problems. (Bloom: Apply)

CO4

Solve first-order differential equations and understand Laplace transforms. (Bloom: Apply)

CO–PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-

Legends: 1–Low; 2–Medium; 3–High; – No Correlation

📋 Curriculum Coverage Map

Module → Topics → Hours → CO → Handbook Anchor

Module	Major Topics	Hours	CO	Section
I	Complex number system, operations, Argand plane, modulus, amplitude, polar form	14	CO1	Module 1
II	Scalars & vectors, addition, subtraction, scalar multiplication, dot/cross product, applications (area, work, moment)	15	CO2	Module 2
III	Integration as reverse, standard integrals, substitution, parts, definite integrals, area under curve	15	CO3	Module 3
IV	Differential equations (order/degree, variable separable, linear), Laplace transforms (elementary functions, shifting property, inverse)	16	CO4	Module 4

🗺️ How the Modules Connect

Module 1

Complex Numbers

Foundation for AC circuit analysis, signal processing, and control systems in electrical engineering.

Module 2

Vector Algebra

Essential for electromagnetics, mechanics, and field analysis — force, torque, work, and moments.

Module 3

Integral Calculus

Area, volume, work, and modelling dynamic systems via accumulation and area under curves.

Module 4

D.E. & Laplace

Modelling electrical circuits, control systems, and signal processing using differential equations and transforms.

MODULE I

Complex Numbers

14 hours (L:10 + T:4) | CO1 | Bloom: Apply

🎯 Learning Goals

- Understand the complex number system
- Perform fundamental operations
- Represent complex numbers graphically
- Find modulus, amplitude, and polar form
- Solve engineering problems

🔑 Key Facts

- $i = \sqrt{(-1)}$
- $z = a + ib$
- $|z| = \sqrt{(a^2 + b^2)}$
- $\arg(z) = \tan^{-1}(b/a)$
- Polar: $z = r(\cos\theta + i\sin\theta)$

The Complex Number System

A **complex number** is a number of the form $\mathbf{z = a + ib}$, where **a** and **b** are real numbers, and **i** is the imaginary unit defined by $\mathbf{i^2 = -1}$.

- **Real part:** $\mathrm{Re}(z) = a$
- **Imaginary part:** $\mathrm{Im}(z) = b$

> $z = a + ib \;\cdot\; a,b \in \mathbb{R} \;\cdot\; i^2 = -1$

Example: $z = 3 + 4i$ has Re = 3 and Im = 4.

> **Malayalam support:** □□□□□□□□□□ □□□□□□□□ (complex numbers) □□□□□□□ a + ib □□□□□□□□□□□□□ □□□□□□□□□□□. □□□□□□ 'i' □□□□□□□ √(−1) □□□. □□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□.

> 💡 **Tip:** The real and imaginary parts are always real numbers. The imaginary part includes the coefficient of i, not i itself.

Conjugate and Equality

Conjugate: The conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

$$\bar{z} = a - ib$$

Equality: Two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal iff $a_1 = a_2$ **and** $b_1 = b_2$.

Example: If $2x + 3iy = 4 + 6i$, then $2x = 4 \Rightarrow x = 2$, and $3y = 6 \Rightarrow y = 2$.

💡 **Tip:** The conjugate is used to rationalise denominators and find modulus.

Fundamental Operations

For $z_1 = a + ib$ and $z_2 = c + id$:

- **Addition:** $z_1 + z_2 = (a+c) + i(b+d)$
- **Subtraction:** $z_1 - z_2 = (a-c) + i(b-d)$
- **Multiplication:** $z_1 \cdot z_2 = (ac - bd) + i(ad + bc)$
- **Division:** $z_1 / z_2 = (a+ib)/(c+id) \times (c-id)/(c-id) = [(ac+bd) + i(bc-ad)] / (c^2 + d^2)$

Example (Multiplication): $(2+3i)(1-4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$

Example (Division): $(3+4i)/(1-2i) = (3+4i)(1+2i)/5 = (3+6i+4i+8i^2)/5 = (-5+10i)/5 = -1 + 2i$

⚠️ **Common mistake:** In division, always multiply numerator and denominator by the conjugate of the denominator.

Graphical Representation — Argand Plane

Complex numbers are represented on the **Argand plane** (complex plane) with the real part on the horizontal axis and the imaginary part on the vertical axis.

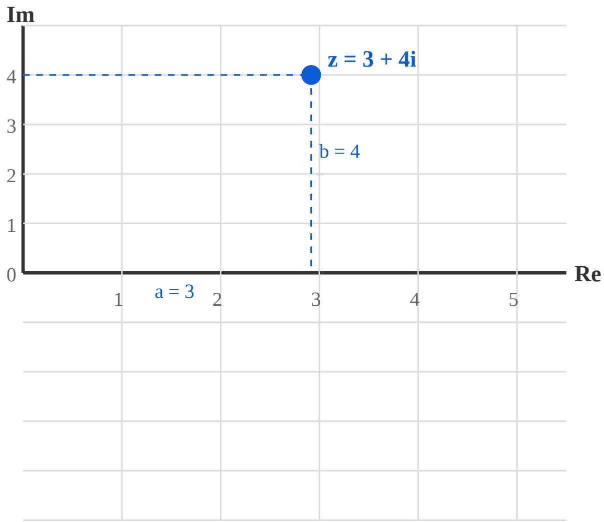

Figure 1: Argand plane showing $z = 3 + 4i$ with real part $a = 3$ and imaginary part $b = 4$.

> **Malayalam support:** Argand plane-□ complex numbers-□□ points □□□ represent □□□□□□□□□□. Horizontal axis real part-um vertical axis imaginary part-um □□□.

Modulus and Amplitude

Modulus (Absolute value): $|z| = \sqrt{a^2 + b^2}$. It is the distance from the origin to the point (a, b).

Amplitude (Argument): $\theta = \arg(z) = \tan^{-1}(b/a)$, measured in degrees (or radians).

$$|z| = \sqrt{a^2 + b^2} \cdot \theta = \tan^{-1}(b/a)$$

Example: For $z = 3 + 4i$: $|z| = \sqrt{9+16} = 5$, $\theta = \tan^{-1}(4/3) \approx 53.13°$

⚠️ **Common mistake:** The argument depends on the quadrant. Always check the signs of a and b. In QII, $\theta = 180° - \tan^{-1}(b/|a|)$.

Polar Form

A complex number can be expressed in **polar form** as:

> $z = r(\cos\theta + i\sin\theta) \;\cdot\; r = |z|, \theta = \arg(z)$

This is also written as **$z = r\angle\theta$** or **$z = r \operatorname{cis} \theta$**.

Example: For $z = 3 + 4i$: $r = 5$, $\theta = 53.13°$ ⇒ $z = 5(\cos 53.13° + i \sin 53.13°)$

> 💡 **Tip:** Polar form is especially useful for multiplication and division — multiply moduli and add arguments.

Solved Problems

Problem 1: Find the modulus and argument of $z = -1 + i$.
Solution: $a = -1$, $b = 1$. $r = \sqrt{(1+1)} = \sqrt{2}$. $\theta = 180° - \tan^{-1}(1/1) = 180° - 45° = 135°$.

Problem 2: Express $z = 2 - 2i$ in polar form.
Solution: $r = \sqrt{(4+4)} = 2\sqrt{2}$. $\theta = -\tan^{-1}(2/2) = -45°$. $z = 2\sqrt{2}(\cos(-45°) + i \sin(-45°))$.

Problem 3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.
Solution: $(1+2i)(3-i) = 3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$.

> 💡 **Engineering Application:** Complex numbers model AC circuits — impedance $Z = R + iX$, with magnitude giving total opposition and phase angle giving phase shift.

🧮 Complex Number Calculator

Enter real and imaginary parts to compute modulus, argument, and polar form.

Real part (a)	Imaginary part (b)
3	4

$z = 3 + 4i$ | $|z| = 5.000$ | arg = 53.13° | Polar: $5.000(\cos 53.13° + i \sin 53.13°)$

MODULE II

Vector Algebra

15 hours (L:12 + T:3) | CO2 | Bloom: Apply

🎯 Learning Goals

- Distinguish scalars and vectors
- Perform addition, subtraction, scalar multiplication
- Compute dot and cross products
- Apply vectors to engineering problems: area, work, moment

🔑 Key Facts

- $a \cdot b = |a||b| \cos \theta$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$
- Work = $F \cdot d$
- Moment = $r \times F$
- Area of parallelogram = $|a \times b|$

Scalars, Vectors, and Unit Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, temperature, speed).
- **Vector:** A quantity with both magnitude and direction (e.g., velocity, force, displacement).

Unit vectors: $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$ along x, y, z axes respectively. Each has magnitude 1.

Position vector: $r = x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}$.

> **Malayalam support:** Scalar-□□ magnitude □□□□□□□ □□□□□ (□□□: mass). Vector-□□ magnitude-□□ direction-□□ □□□□□ (□□□: velocity). Unit vectors î, ĵ, k̂ axes-□□ □□□□□□□□□□□□□□□□□□□.

Magnitude and Unit Vector

Magnitude: $|v| = \sqrt{x^2 + y^2 + z^2}$ for $v = x\,\hat{\imath} + y\,\hat{\jmath} + z\,\hat{k}$.

Unit vector: $\hat{v} = v / |v|$ (direction of v).

Example: `v = 3î + 4ĵ. |v| = 5. v̂ = (3/5)î + (4/5)ĵ.`

Vector Addition, Subtraction, Scalar Multiplication

- **Addition:** $v + w = (v_1+w_1)\hat{\imath} + (v_2+w_2)\hat{\jmath} + (v_3+w_3)\hat{k}$
- **Subtraction:** $v - w = (v_1-w_1)\hat{\imath} + (v_2-w_2)\hat{\jmath} + (v_3-w_3)\hat{k}$
- **Scalar multiplication:** $c\cdot v = (c\,v_1)\hat{\imath} + (c\,v_2)\hat{\jmath} + (c\,v_3)\hat{k}$

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v+w = 3î+1ĵ, v−w = 1î+5ĵ, 3v = 6î+9ĵ.`

Dot Product

$$v \cdot w = |v||w| \cos\theta \quad \cdot \quad v \cdot w = v_1 w_1 + v_2 w_2 + v_3 w_3$$

- **Angle:** $\cos\theta = (v \cdot w)/(|v||w|)$
- **Perpendicular:** $v \perp w \Leftrightarrow v \cdot w = 0$
- **Work done:** $W = F \cdot d$ (force × displacement)

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v·w = 2(1)+3(−2) = 2−6 = −4. |v|=√13, |w|=√5, cos θ = −4/(√13·√5) ≈ −0.496.`

Cross Product

$$v \times w = |v||w| \sin\theta \cdot \hat{n} \quad \cdot \quad v \times w = (v_2 w_3 - v_3 w_2)\hat{\imath} - (v_1 w_3 - v_3 w_1)\hat{\jmath} + (v_1 w_2 - v_2 w_1)\hat{k}$$

- **Parallel:** $v \parallel w \Leftrightarrow v \times w = 0$
- **Area of parallelogram:** $|v \times w|$
- **Area of triangle:** $\frac{1}{2} |v \times w|$
- **Moment (torque):** $\tau = r \times F$

Example: `v = 2î+3ĵ, w = 1î−2ĵ. v×w = (3·0−0·(−2))î − (2·0−0·1)ĵ + (2·(−2)−3·1)k̂ = −7k̂.`

Applications: Area, Work, Moment

Area of parallelogram: Vectors $a = 2\hat{i}+3\hat{j}$, $b = 1\hat{i}-2\hat{j}$. $|a\times b| = |-7\hat{k}| = 7$ sq. units.

Area of triangle: $\frac{1}{2} \times 7 = 3.5$ sq. units.

Work done: Force $F = 3\hat{i}+4\hat{j}$ N, displacement $d = 2\hat{i}+1\hat{j}$ m. $W = F\cdot d = 3(2)+4(1) = 10$ J.

Moment: Position $r = 2\hat{i}+3\hat{j}$, Force $F = 1\hat{i}-2\hat{j}$. $\tau = r\times F = (3\cdot0-0\cdot(-2))\hat{i} - (2\cdot0-0\cdot1)\hat{j} + (2\cdot(-2)-3\cdot1)\hat{k} = -7\hat{k}$ N·m.

💡 **Engineering Application:** Vectors are essential in electromagnetics (electric/magnetic fields), mechanics (forces, torques), and structural analysis.

🧮 Vector Calculator

Enter components of two vectors (a, b) and compute dot product, cross product, magnitudes, angle, area, and work (with force and displacement).

a_1	a_2
2	3
a_3	b_1
0	1
b_2	b_3
-2	0

a·b = -4.000 | a×b = (0.000, 0.000, -7.000) | |a| = 3.606 | |b| = 2.236 | θ = 119.74° | Area = 7.000 | Work (a as force, b as displacement) = -4.000

MODULE III

Integral Calculus

15 hours (L:11 + T:4) | CO3 | Bloom: Apply

🎯 Learning Goals

- Understand integration as reverse of differentiation
- Recall standard integrals
- Apply substitution and integration by parts
- Evaluate definite integrals
- Find area bounded by a curve and x-axis

🔑 Key Facts

- $\int x^n \, dx = x^{n+1}/(n+1) + C$ $(n \neq -1)$
- $\int e^x \, dx = e^x + C$
- $\int 1/x \, dx = \ln|x| + C$
- $\int \sin x \, dx = -\cos x + C$
- $\int \cos x \, dx = \sin x + C$

Integration as Reverse of Differentiation

Integration is the inverse process of differentiation. If $d/dx\,[F(x)] = f(x)$, then $\int f(x)\,dx = F(x) + C$, where C is the constant of integration.

$$\int f(x)\,dx = F(x) + C \quad \cdot \quad d/dx\,[F(x)] = f(x)$$

Example: $d/dx\,(x^2) = 2x \Rightarrow \int 2x\,dx = x^2 + C$.

💡 **Tip:** The constant C accounts for all antiderivatives. Always include it in indefinite integrals.

Standard Integrals

$\int x^n \, dx = x^{n+1}/(n+1) + C$

$n \neq -1$

$\int 1/x \, dx = \ln|x| + C$

$x \neq 0$

$\int e^x \, dx = e^x + C$

$\int \sin x \, dx = -\cos x + C$

$\int \cos x \, dx = \sin x + C$

$\int \sec^2 x \, dx = \tan x + C$

$\int \mathrm{cosec}^2 x \, dx = -\cot x + C$

$\int \sec x \tan x \, dx = \sec x + C$

$\int \mathrm{cosec}\, x \cot x \, dx = -\mathrm{cosec}\, x + C$

Rules of Integration & Substitution

Rules:

- $\int k \cdot f(x)\,dx = k \cdot \int f(x)\,dx$
- $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

Substitution:

- $\int f(ax+b)\,dx = (1/a)\,F(ax+b) + C$
- $\int f(x) \cdot f'(x)\,dx = [f(x)]^2/2 + C$
- $\int f'(x)/f(x)\,dx = \ln|f(x)| + C$

Example (substitution): $\int (2x+3)^5\,dx$. Let $u = 2x+3$, $du = 2dx \Rightarrow dx = du/2$. $= 1/2 \int u^5\,du = u^6/12 + C = (2x+3)^6/12 + C$.

Example (f'/f): $\int (2x)/(x^2+1)\,dx = \ln|x^2+1| + C$.

Integration by Parts

For product of two functions:

> $\int u\,dv = uv - \int v\,du$

LIATE rule (choose u as the function that comes first): Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

Example: $\int x \sin x\,dx$. Let $u = x$ (algebraic), $dv = \sin x\,dx \Rightarrow du = dx$, $v = -\cos x$. $= x(-\cos x) - \int (-\cos x)\,dx = -x\cos x + \sin x + C$.

> ⚠️ **Common mistake:** Choosing the wrong u can make the integral more complex. Use LIATE.

Definite Integrals

A **definite integral** has limits of integration:

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

Example: $\int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3 - 0 = 1/3$.

Area Bounded by a Curve and the x-axis

The area enclosed by the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is:

$$\text{Area} = \int_a^b f(x)\,dx \quad \cdot \quad (\text{if } f(x) \geq 0 \text{ on } [a,b])$$

Example: Area under $y = x^2$ from $x=0$ to $x=2$ is $\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$ sq. units.

💡 **Engineering Application:** Definite integrals compute areas, volumes, work done, centre of mass, and signal energy in electrical engineering.

🧮 Definite Integral Calculator (polynomial)

Compute $\int_a^b x^n\,dx$. Enter a, b, and n.

a (lower limit)

0

b (upper limit)

2

n (exponent)

2

$\int_a^b x^n\,dx = 2.666667$ (a=0, b=2, n=2)

MODULE IV

Differential Equations & Laplace Transforms

16 hours (L:12 + T:4) | CO4 | Bloom: Apply

🎯 Learning Goals

- Identify order and degree of differential equations
- Solve first-order variable separable and linear D.E.
- Understand Laplace transforms of elementary functions
- Apply shifting property
- Evaluate inverse Laplace transforms

🔑 Key Facts

- $L\{1\} = 1/s$
- $L\{t^n\} = n!/s^{n+1}$
- $L\{e^{at}\} = 1/(s-a)$
- $L\{\sin at\} = a/(s^2+a^2)$
- $L\{\cos at\} = s/(s^2+a^2)$

Differential Equations — Order and Degree

A **differential equation** is an equation involving derivatives of a function.

- **Order:** The highest derivative present.
- **Degree:** The exponent of the highest derivative (after removing radicals and fractions).

Example: $d^2y/dx^2 + 3(dy/dx) + 2y = 0$ → Order 2, Degree 1.

Example: $(d^2y/dx^2)^3 + (dy/dx)^2 + y = 0$ → Order 2, Degree 3.

⚠️ **Common mistake:** Degree is only defined when the equation is polynomial in derivatives. If derivatives appear in denominators or under radicals, rationalise first.

First-Order Differential Equations — Variable Separable

Form: $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x)\,dx$

Example: $dy/dx = xy \Rightarrow dy/y = x\,dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = C\,e^{(x^2/2)}$.

💡 **Tip:** Separate variables so that all y terms are on one side and all x terms on the other.

First-Order Linear Differential Equations

Form: $dy/dx + P(x)y = Q(x)$

Integrating factor: I.F. = $e^{(\int P\,dx)}$ · $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Example: $dy/dx + 2y = x$. $P = 2$, I.F. = $e^{(\int 2\,dx)} = e^{(2x)}$. $y\,e^{(2x)} = \int x\,e^{(2x)}\,dx + C$. Using integration by parts: $\int x\,e^{(2x)}\,dx = (x/2)e^{(2x)} - (1/4)e^{(2x)}$. So $y = (x/2) - (1/4) + C\,e^{(-2x)}$.

Introduction to Laplace Transforms

The **Laplace transform** of a function f(t) is defined as:

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s)$$

It transforms a function of time t into a function of complex frequency s.

Malayalam support: Laplace transform □□□□□□ time domain-□ □□□□□□ s-domain (frequency domain) □□□□□□ □□□□□□□□□ □□□ □□□□ □□□□□□□□□. □□□ differential equations-□□ algebraic equations-□□□□□ □□□□□□□□□□.

💡 **Engineering Application:** Laplace transforms are fundamental in control systems, circuit analysis, and signal processing.

Laplace Transforms of Elementary Functions

$L\{1\} = 1/s$

$(s > 0)$

$L\{t^n\} = n! / s^{n+1}$

$n = 0,1,2,\ldots$

$L\{e^{at}\} = 1/(s-a)$

$(s > a)$

$L\{\sin at\} = a/(s^2+a^2)$

$L\{\cos at\} = s/(s^2+a^2)$

$L\{\sinh at\} = a/(s^2-a^2)$

$L\{\cosh at\} = s/(s^2-a^2)$

Linearity property: $L\{a f(t) + b g(t)\} = a L\{f(t)\} + b L\{g(t)\}$

Example: $L\{3 + 2e^{4t}\} = 3/s + 2/(s-4)$.

Shifting Property (First Shifting Theorem)

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Example: $\mathcal{L}\{e^{3t} \sin 2t\} = 2 / ((s-3)^2 + 4)$.

Example: $\mathcal{L}\{e^{-2t} \cos 5t\} = (s+2) / ((s+2)^2 + 25)$.

⚠️ **Common mistake:** The shifting property shifts s by $-a$ for $\mathcal{L}\{e^{at} f(t)\}$. Remember the sign: $e^{at} \to F(s-a)$.

Inverse Laplace Transforms

The inverse Laplace transform recovers $f(t)$ from $F(s)$. Common inverses:

$L^{-1}\{1/s\} = 1$

$L^{-1}\{1/s^2\} = t$

$L^{-1}\{1/(s-a)\} = e^{at}$

$L^{-1}\{1/(s^2+a^2)\} = (1/a) \sin at$

$L^{-1}\{s/(s^2+a^2)\} = \cos at$

Example: $L^{-1}\{1/(s^2+9)\} = (1/3) \sin 3t$.

Example: $L^{-1}\{s/(s^2+16)\} = \cos 4t$.

💡 **Tip:** Use partial fractions to simplify $F(s)$ before finding inverse transforms.

🧮 Laplace Transform Helper

Select a function to see its Laplace transform and inverse.

Function f(t)

1

a

2

b (if needed)

3

$L\{1\} = 1/s$ | $L^{-1}\{1/s\} = 1$

📐 Formula Bank

$z = a + ib,\ i^2 = -1$

Complex number

$|z| = \sqrt{a^2 + b^2}$

Modulus

$\theta = \tan^{-1}(b/a)$

Argument

$z = r(\cos\theta + i\sin\theta)$

Polar form

$a \cdot b = |a||b| \cos\theta$

Dot product

$a \times b = |a||b| \sin\theta \cdot \hat{n}$

Cross product

$\int x^n\,dx = x^{n+1}/(n+1) + C$

Power rule

$\int u\,dv = uv - \int v\,du$

Integration by parts

$\int_a^b f(x)\,dx = F(b) - F(a)$

Definite integral

$\text{I.F.} = e^{\int P\,dx}$

Linear D.E.

$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$

Laplace transform

$L\{e^{at} f(t)\} = F(s-a)$

Shifting property

🖼️ Diagram Library for Rapid Revision

📐 Argand Plane

Module 1 — Argand Plane showing complex number representation.

📐 Vector Geometry

Visualise dot and cross products in 3D space. Module 2

📈 Area Under Curve

Visualise area bounded by a curve and x-axis. Module 3

📝 Self-Learning Activities (Official)

Week 1-4 · 16 h

Module I — Complex Numbers

- Find conjugate, modulus, amplitude, polar form
- Problems on equality, addition, subtraction
- Multiply and divide complex numbers
- Applications in engineering areas

Week 5-8 · 15 h

Module II — Vector Algebra

- Identify scalars and vectors
- Vector addition, subtraction, scalar multiplication
- Dot product and cross product problems
- Work done and moment problems
- Applications of vectors

Week 9-12 · 14 h

Module III — Integral Calculus

- Find integrals of functions
- Evaluate definite integrals
- Area bounded by curve and x-axis
- Applications of integral calculus

Week 13-15 · 15 h

Module IV — D.E. & Laplace

- Solve first-order differential equations
- Problems using first shifting property
- Evaluate Laplace transforms of functions
- Applications of D.E. and Laplace

Total self-learning hours: 60 (as per syllabus)

📊 Assessment Methodology

CIE / ESE Breakup

Mode	Attendance	CA1	CA2	CA3	CA4	Series Exam (2 modules)	Series Exam (2 modules)	ESE
Duration	—	2h	2h	2h	2h	2h	2h	2.5h
Exam mark	—	15	15	15	15	15	15	60
Converted to	5	15				20		60
Total	40 (CIE) + 60 (ESE)							

Series Examination Pattern

Duration: 2 hours
Total marks: 50 (converted to 20)

- Part A: 2 × 1 mark
- Part B: 2 × 3 marks
- Part C: 2 × 7 marks (with choice)

End Semester Examination

Duration: 2.5 hours
Total marks: 60

- Part A: 1 mark × 12 questions (2 per module)
- Part B: 3 marks × 6 questions (2 out of 3 per module)
- Part C: 7 marks × 6 questions (1 set with choice per module)

❓ Module-wise Questions with Answers

Module I — Complex Numbers

Q1: Find the modulus and argument of $z = 1 + i$.▼

$|z| = \sqrt{(1^2+1^2)} = \sqrt{2}$. $\theta = \tan^{-1}(1/1) = 45°$.

Q2: Express $z = 2 - 2i$ in polar form.▼

$r = \sqrt{(4+4)} = 2\sqrt{2}$. $\theta = -45°$. $z = 2\sqrt{2}(\cos(-45°) + i \sin(-45°))$.

Q3: If $z_1 = 1+2i$ and $z_2 = 3-i$, find $z_1 \times z_2$.▼

$(1+2i)(3-i) = 3-i+6i-2i^2 = 3+5i+2 = 5+5i$.

Module II — Vector Algebra

Q1: Find the magnitude of $v = 3\hat{\imath} + 4\hat{\jmath}$.▼

$|v| = \sqrt{(9+16)} = 5$.

Q2: Compute the dot product of a = (2,3) and b = (1,−2).▼

$a \cdot b = 2(1)+3(-2) = 2-6 = -4$.

Q3: Find the cross product of $a = 2\hat{\imath}+3\hat{\jmath}$ and $b = 1\hat{\imath}-2\hat{\jmath}$.▼

$a\times b = (3\cdot0-0\cdot(-2))\hat{\imath} - (2\cdot0-0\cdot1)\hat{\jmath} + (2\cdot(-2)-3\cdot1)\hat{k} = -7\hat{k}$.

Q4: What is the area of a parallelogram formed by vectors a and b from Q3?▼

Area = $|a\times b|$ = 7 sq. units.

Module III — Integral Calculus

Q1: Evaluate $\int 3x^2\,dx$.▼

$\int 3x^2\,dx = 3\cdot(x^3/3) + C = x^3 + C$.

Q2: Evaluate $\int_0^1 x\,dx$.▼

$[x^2/2]_0^1 = 1/2 - 0 = 1/2$.

Q3: Find the area under $y = x^2$ from x=0 to x=2.▼

$\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$ sq. units.

Module IV — Differential Equations & Laplace

Q1: Find the order and degree of $d^2y/dx^2 + 3(dy/dx) + 2y = 0$.▼

Order = 2, Degree = 1.

Q2: Solve $dy/dx = 2xy$.▼

$dy/y = 2x\,dx \Rightarrow \ln|y| = x^2 + C \Rightarrow y = C e^{(x^2)}$.

Q3: Find L{sin 3t}.▼

$L\{\sin 3t\} = 3/(s^2+9)$.

Q4: Find $L^{-1}\{1/(s^2+25)\}$.▼

$L^{-1}\{1/(s^2+25)\} = (1/5) \sin 5t$.

📄 Practice Model Paper

Note: This is a practice paper based on the official pattern, not an official SITTTR model question paper.

Course 2001B — Mathematics for Electrical Sciences

Duration: 2.5 hours · **Total marks:** 60

All modules are covered. Answer all parts.

Part A (1 mark each — 12 questions, 2 per module)

- **Q1.** What is the conjugate of 2 + 5i? (M1)
- **Q2.** Find |3 − 4i|. (M1)
- **Q3.** What is a unit vector? (M2)
- **Q4.** When are two vectors perpendicular? (M2)
- **Q5.** $\int 2x\,dx$ = ? (M3)
- **Q6.** What is the order of $d^2y/dx^2 + y = 0$? (M3)
- **Q7.** Define Laplace transform. (M4)
- **Q8.** Find L{1}. (M4)
- **Q9-12.** (Additional questions covering all modules)

Part B (3 marks each — 6 questions, 2 out of 3 per module)

- **M1:** Express $1 - i$ in polar form. *OR* Find modulus and argument of $-1 + i$.
- **M2:** Given $a = (1,2,0)$, $b = (0,1,1)$, find $a \cdot b$ and $a \times b$. *OR* Find work done if $F=2\hat{i}+3\hat{j}$ and $d=1\hat{i}-2\hat{j}$.
- **M3:** Evaluate $\int x \sin x \, dx$. *OR* Find area under $y = 2x$ from $x=0$ to $x=3$.
- **M4:** Solve $dy/dx + 2y = x$. *OR* Find L{e^{3t} sin 2t}.

Part C (7 marks each — 6 questions, 1 set with choice per module)

- **M1:** If $z_1 = 2+3i$ and $z_2 = 1-2i$, find z_1+z_2, z_1-z_2, $z_1 \cdot z_2$, and z_1/z_2. *OR* Express $z = 2+2i$ in polar form and find z^3.
- **M2:** For vectors $a = 2\hat{i}+\hat{j}-\hat{k}$ and $b = \hat{i}-2\hat{j}+3\hat{k}$, find $a \cdot b$, $a \times b$, area of parallelogram, and angle between them. *OR* Find moment $\tau = r \times F$ for $r=3\hat{i}+2\hat{j}$, $F=1\hat{i}-4\hat{j}$.
- **M3:** Evaluate $\int_0^2 (x^2+2x) \, dx$. Find area bounded by $y = x^2$ and x-axis from $x=0$ to $x=2$. *OR* Evaluate $\int x^2 e^x \, dx$.
- **M4:** Solve $dy/dx + 2y = e^x$. Find L{e^{−2t} cos 3t}. *OR* Find $L^{-1}\{1/(s^2+4s+13)\}$ (using completing square and shifting).

⚡ Quick Revision

Module I — Complex Numbers

- $z = a + ib$, $i^2 = -1$
- $|z| = \sqrt{(a^2+b^2)}$, $\theta = \tan^{-1}(b/a)$
- Polar: $z = r(\cos\theta + i\sin\theta)$
- Operations: add real/imag separately

Module II — Vector Algebra

- $a \cdot b = |a||b| \cos\theta = a_1 b_1 + a_2 b_2 + a_3 b_3$
- $a \times b = |a||b| \sin\theta \cdot \hat{n}$
- Area = $|a \times b|$, Work = $F \cdot d$, Moment = $r \times F$

Module III — Integral Calculus

- $\int x^n \, dx = x^{n+1}/(n+1) + C$
- $\int u \, dv = uv - \int v \, du$
- Definite: $\int_a^b = F(b) - F(a)$
- Area = $\int_a^b f(x) \, dx$

Module IV — D.E. & Laplace

- I.F. = $e^{\int P \, dx}$ for linear D.E.
- $L\{1\} = 1/s$, $L\{e^{at}\} = 1/(s-a)$
- $L\{\sin at\} = a/(s^2+a^2)$, $L\{\cos at\} = s/(s^2+a^2)$
- $L\{e^{at} f(t)\} = F(s-a)$

⚠️ Common Mistakes — Exam Checklist

- ✔️ In complex division, multiply by conjugate of denominator
- ✔️ In vectors, cross product gives a vector, dot product gives a scalar
- ✔️ In integration, always add +C for indefinite integrals
- ✔️ In Laplace, shifting property: $L\{e^{at} f(t)\} = F(s-a)$ (note the sign)
- ✔️ In differential equations, identify order and degree correctly
- ✔️ In inverse Laplace, use partial fractions when needed

📖 Glossary

Term	Meaning
Complex number	$z = a + ib$, where $i^2 = -1$
Modulus	$\lvert z\rvert = \sqrt{a^2+b^2}$, distance from origin
Argument	$\theta = \tan^{-1}(b/a)$, angle with real axis
Polar form	$z = r(\cos\theta + i\sin\theta)$
Scalar	Quantity with only magnitude
Vector	Quantity with magnitude and direction
Dot product	Scalar product: $a\cdot b = \lvert a\rvert\lvert b\rvert\cos\theta$
Cross product	Vector product: $a\times b = \lvert a\rvert\lvert b\rvert\sin\theta\cdot\hat{n}$
Integration	Reverse process of differentiation
Definite integral	$\int_a^b f(x)\,dx = F(b) - F(a)$
Differential equation	Equation involving derivatives
Laplace transform	$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$
Shifting property	$L\{e^{at}f(t)\} = F(s-a)$
Inverse Laplace	Recovers $f(t)$ from $F(s)$
Integrating factor	I.F. $= e^{\int P\,dx}$ for linear D.E.

📚 Learning Resources

Textbooks

Sl. No.	Title	Author / Publication
1	Mathematics for Electrical Sciences	SITTTR Kalamasser

References

Sl. No.	Title	Author / Publication
1	Schaum's outline of Complex Variables	Murray R. Spiegel, Seymour Lipschutz, John J. Schiller, McGraw Hill
2	Schaum's outline of Vector Analysis	Murray R. Spiegel, McGraw Hill
3	Advanced Engineering Mathematics (10th Ed.)	Erwin Kreyszig, Wiley & Sons
4	A Textbook of Engineering Mathematics	N.P. Bali, Manish Goyal, University Science Press
5	Advanced Engineering Mathematics	B. S. Grewal, Mercury Learning & Information

Web resources: None officially listed in the syllabus.